MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5787WA

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Voltage regulator, subminiature

Outline --- See figure 1

Cathode --- Glow discharge

Base connections:

Pin No.	---	1	3	5
Element	---	k	a	k

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Total darkness ionization voltage	Ambient light ionization voltage	Operating voltage	Operating current	TE	TA	(E) Alt
Unit:	Vdc	Vdc	Vdc	mAdc	°C	°C	ft
Maximum:	---	---	98	25	225	150	Note 1
Minimum:	170	141	---	5	---	-55	---
TEST CONDITIONS:	---	---	---	---	---	---	---

GENERAL:

Qualification - Required

Reliable tube

(E) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3347	Ionization voltage (1)	2	Ebb Ib = 5 to 25 mAdc; illumination = 5 to 50 ftC	0.4	II	Ez	---	135	Vdc
3337	Voltage drop (1)	-	Ebb Ib = 25 mAdc	0.4	II	Etd	95	103	Vdc
3337	Voltage drop (2)	-	Ebb Ib = 5 mAdc	0.4	II	Etd	95	103	Vdc
3335	Regulation	-	Etd (1) - Etd (2)	0.4	II	Reg	---	±3	Vdc
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
3278	Noise and oscillation	-	Ebb Ib = 25 mAdc	1.0	I	Ep	---	20	mVac
3278	Oscillation test	-	Esig = 100 mVac; Rf = 500 ohms; Ebb Ib = 5 to 25 mAdc	1.0	I	---	---	---	---
3347	Ionization voltage (2)	3		6.5	Code G	Ez	---	170	Vdc
3305	Leakage current	-	Eb = 50 Vdc; Rp = 3,000 ohms	6.5	Code G	LIb	---	5	μAdc
---	Voltage repeatability	4	Ebb Ib = 10 mAdc	6.5	Code E	ΔEtd	---	1.0	Vdc
1031	Low-frequency vibration	-	Rp = 10,000 ohms; Ebb Ib = 25 mAdc; F = 40 Hz; 15G	6.5	Code G	Ep	---	50	mVac
1116	Lead fatigue	-		2.5	Code G	---	---	---	---
1041	Shock	-	450 G	---	---	---	---	---	---
1031	Vibration fatigue	5	2.5G; fixed frequency; F = 25 (min), 60 (max)	6.5	---	---	---	---	---
---	Post-shock and vibration-fatigue test end points:								
1031	Low-frequency vibration	-		---	---	Ep	---	100	mVac
3347	Ionization voltage (1)	-		---	---	Ez	---	141	Vdc
3337	Voltage drop (1)	-		---	---	Etd	95	105	Vdc
3337	Voltage drop (2)	-		---	---	Etd	95	105	Vdc
3335	Regulation	-		---	---	Reg	---	±4.0	Vdc
[illegible]	Glass strain	-		2.5	I	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
						1ST SAMPLE	COMBINED SAMPLES		MIN	MAX	
	Quality conformance inspection, part 3										
1516	Stability life (1 hour)	-	Ebb Ib = 25 mAdc; TA = room	6.5	Code H	---	---	---	---	---	---
---	Stability life-test end points:										
3337	Change in voltage drop (1) of individual tubes	-		---	---	---	---	ΔEtd t	---	1.0	Vdc
3337	Change in voltage drop (2) of individual tubes	-		---	---	---	---	ΔEtd t	---	1.0	Vdc
1521	Survival-rate life	-	Stability life-test, or equivalent conditions	---	II	---	---	---	---	---	---
---	Survival-rate life-test end points:										
1201	Short and discontinuity detection	-		0.65	---	---	---	---	---	---	---
3337	Change in voltage drop (1) of individual tubes	-		6.5	---	---	---	ΔEtd t	---	1.0	Vdc
3337	Change in voltage drop (2) of individual tubes	-		6.5	---	---	---	ΔEtd t	---	1.0	Vdc
1501	Intermittent life	6	Stability life-test conditions; TE = 225° C (min)	---	---	---	---	---	---	---	---
---	Intermittent life-test end points (500 hours):										
---	Inoperatives	-		---	---	1	3	---	---	---	---
3335	Regulation	-		---	---	1	3	Reg	---	± 5.0	Vdc
3337	Voltage drop (1)	-		---	---	1	3	Etd	95	105	Vdc
3337	Voltage drop (2)	-		---	---	1	3	Etd	95	105	Vdc
3337	Change in voltage drop (1) of individual tubes	-		---	---	1	3	ΔEtd t	---	3	%
3337	Change in voltage drop (2) of individual tubes	-		---	---	1	3	ΔEtd t	---	3	%
3347	Ionization voltage (1)	-		---	---	1	3	Ez	---	141	Vdc
---	Total defectives	-		---	---	4	8	---	---	---	---
---	Intermittent life-test end points (1,000 hours):										
---	Inoperatives	-		---	---	2	5	---	---	---	---
3335	Regulation	-		---	---	2	5	Reg	---	± 6.0	Vdc
3337	Voltage drop (1)	-		---	---	2	5	Etd	95	105	Vdc
3337	Voltage drop (2)	-		---	---	2	5	Etd	95	105	Vdc
3337	Change in voltage drop (1) of individual tubes	-		---	---	2	5	ΔEtd t	---	4	%
3337	Change in voltage drop (2) of individual tubes	-		---	---	2	5	ΔEtd t	---	4	%
3347	Ionization voltage (1)	-		---	---	2	5	Ez	---	141	Vdc
---	Total defectives	-		---	---	5	10	---	---	---	---

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NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.

2. This test to be performed at the conclusion of the holding period.

3. Conditions for this test shall be those of ionization voltage (1), except testing shall be done in total darkness and the tube shall not have conducted or have been exposed to light for at least 24 hours prior to testing.

4. Repeatability shall be defined as the maximum shift in voltage drop between successive firings of the tube. The tube shall be tested in the following manner:

   (a) Etd shall be read at 10 mAdc drain.

   (b) The tube shall be turned off for 1 minute.

   (c) The tube shall be restarted and operated at the same current.

   (d) Etd shall be read after 1 minute of operation.

   (e) The on-off cycle shall be repeated a minimum of five times.

(E) 5. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When a lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

(E) 6. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Etd (± 3 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES
ARMY - MI

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2607)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-26254

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